

LungVision AI — Project Documentation

Title: *LungVision AI: Multimodal Deep Learning for Lung Cancer Detection*

Developed by: *Junaid Shariff and team*

Introduction

LungVision AI is a deep learning-based web application for early detection of lung cancer using CT scans and chest X-rays. It combines the power of **CNN's** and **Ensembling** with **Grad-CAM** explainable AI to deliver accurate and interpretable results for medical professionals and researchers.

Objectives

- Detect lung cancer from medical images and csv data (CT scans and X-rays).
 - Provide visual explanations using Grad-CAM.
 - Build a user-friendly web interface for predictions.
 - Achieve high accuracy using transfer learning.
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System Overview

The system follows this pipeline:

1. **Image Upload**
 2. **Preprocessing** (resize, normalize)
 3. **Model Inference** (CNN's [4 for each modelities])
 4. **Classification** (Normal / Malignant)
 5. **Explainability** (Grad-CAM heatmap)
 6. **Ensembling** (Averging 4 models)
 7. **Result Display** (prediction + confidence + visualization)
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Dataset Information:

- **CT Scans:** Custom Dataset form NSCLC
 - **X-rays:** Custom Dataset from NIH Chest X-ray Dataset
 - **Clinical Metadata :** From Kaggle
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Technologies Used

- **Language:** Python
 - **Frameworks:** PyTorch, Flask
 - **Model:** Efficient net B0, Resnet50, Inception V3, Densenet121(Transfer Learning)
 - **Visualization:** Grad-CAM, Matplotlib
 - **Frontend:** HTML, CSS, JavaScript (or React)
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Explainability - Grad-CAM

We use **Gradient-weighted Class Activation Mapping (Grad-CAM)** to highlight regions of the image that influence model predictions.

- CT & X-ray scans return heatmaps overlayed on the input image.
 - Helps radiologists trust and understand the AI's decision.
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How to Use

► Web Application Steps:

1. Visit the website.
 2. Upload a CT scan or chest X-ray image or clinical metadata
 3. View prediction with confidence score.
 4. Analyse heatmap to understand prediction region.
 5. Download or screenshot result if needed.
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Future Work

- Support for full DICOM viewer and 3D scan input.
 - Mobile app for doctors in rural areas.
 - Integration with real-time hospital databases.
 - Add multilingual support and chatbot integration.
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Credits

- **Developer:** Junaid Shariff , Mohammed Adan , Punith J and Md Yousuf
- **Guide:** Dr. Victor A I
- **Special Thanks:** OpenAI, TCIA, Kaggle, PyTorch community